



DELTA X

AUTONOMOUS OPTIONS TRADING · CODE · RISK · EXECUTE

An options agent whose deliverable is not a P&L number. It is the record of every trade it refused, and why.

GOLDEN RULES

56

each one earned from a test

TESTS PASSING

366

0 failing

COMMITTS

81

full rule provenance

BUILD DAYS

BUILD TRACKER

Three days. Fifty-six rules.

Every rule below was written because a test contradicted an assumption we already held. None was designed in advance – the corpus is a record of being wrong in a specific, correctable way.

DAY 01

29 AUG

FOUNDATIONS • GATES • LEDGER • FIRST BACKTESTS

41

RULES SET

E1-E15 • options • stocks • crypto

49

COMMITTS



13 GATES

HASH LEDGER

CHAIN SCREENER

EXECUTION BARRIER

SEC EARNINGS

RSS FEEDS

DAY 02

30 AUG

VALIDATION • THE DAY ASSUMPTIONS BROKE

11

RULES SET
E16-E25 · C-E26

25 COMMITS



PERMISSION STATE

DTE SWEEP

WIRING AUDIT

RISK 10%→30%

KILL SWITCH

VENUE SCREEN

DAY 03

31 AUG

THE NIGHT BEFORE LIVE · HARDENING

4 RULES SET
E27-E30

7 COMMITS · pre-dawn



AUTONOMOUS ON

EXITS BUILT

RECONCILIATION

KEYS ROTATED

PRE-MARKET INTEL

DAY	WHAT BROKE	RULE
29 Aug	Weekend quotes: 122-184% spreads, negative credits	E13
29 Aug	Condor measured negative – until the exit was modelled	E15
30 Aug	Our headline number used an exit that fires after the deadline	E17
30 Aug	5, 7 and 10 DTE do not exist on the chain	E18
30 Aug	A gate passed its unit test and could never fire	E20
30 Aug	TRON: clean history, delisted, unbuyable	E25
31 Aug	The exit rule was documented, cited – and never called	E29

DAY

WHAT BROKE

RULE

31 Aug

Every cycle believed the book was empty; the cap reset every 5 min

E30

THE CORPUS

Fifty-six rules, four domains

Rules are versioned in the repository alongside the code they govern, so every decision the agent logs can be traced to the commit that authorised it.

EXECUTION

29

E1-E30 · gates, timing, validation

STOCKS

12

sizing, selection, expectancy

OPTIONS

10

structure, strikes, premium

CRYPTO

5

venue limits, C-E26

E4	Triggers nominate; gates decide. AI may propose – deterministic code authorises.
E10	Classify before encoding. Nothing gates a trade until it has been backtested.
E16	A global permission state sits above strategy. No candidate can override it.
E19	Per-trade expectancy is not one-week expectancy. The contest gets one draw.
E23	Quote two bounds, not one number. Our estimates differ by 19×; both are stated.
E30	Ask the broker what you hold before opening anything. A cap that resets every cycle is not a cap.

STRATEGY

Sell time, not direction

Five directional strategies were tested and all five failed. The edge that survived is structural: implied volatility is systematically priced above realised. An iron condor harvests that spread without forecasting anything.

STRUCTURE

Instrument	Iron condor – defined risk
Expiry	11 DTE · 11 Sep
Strike delta	0.20
Exit	GTC at 50% credit, rested at fill · 2-DTE time stop
Direction	Put = LONG · Call = SHORT

Universe

17 names, 34 verified tradeable

BOOK

Options

\$30,000 risk

validated

Stocks

\$60,000

covered calls

Crypto

\$10,000

no edge found

The options sleeve produced 83% of backtested return on 30% of capital. The other two are disclosed as what they are.

RISK

Unknown is never safe

Thirteen deterministic gates, every one fail-closed. A missing input is refused, not assumed. Above them sits a permission state no strategy can override.

PER POSITION

2%

\$2,000 max loss

PORTFOLIO

30%

\$30,000 · forces 15+ positions

DAILY KILL SWITCH

-5%

one bad session → HALT

STATE	PUT SPREADS	CALL SPREADS	SIZE	TRIGGER
NORMAL	yes	yes	100%	all clear
CAUTION	yes	yes	50%	any reading unavailable
DEFENSIVE	no	yes	50%	VIX +15% · all benchmarks weak
NO NEW POSITIONS	no	no	0%	VIX +30% · market closed
HALT	no	no	0%	stale data · -5% day · -20% drawdown

EVIDENCE

Three weeks, replayed

Real historical prices, credit modelled at the gate's **minimum** – a real fill does better; no commissions charged – a real fill does worse. Stated in both directions.

WEEK	OPTIONS	STOCKS	CRYPTO	TOTAL
10–14 Aug	+3,737	-279	-77	+3,382
17–21 Aug	-1,728	+494	+3,037	+1,803
24–28 Aug	+2,998	+322	-110	+3,210
TOTAL	+5,007	+537	+2,850	+8,394 +8.39%

The crypto column is almost entirely one week of XRP. **It is not repeatable**, and it is reported here rather than quietly absorbed into a headline.

WALK-FORWARD E · SPY

+0.109

out-of-sample, 2024→

WALK-FORWARD E · IWM

+0.147

out-of-sample, 2024→

POSITIONS PROFITABLE

67%

54 of 81 across 3 weeks

WHAT WE REJECTED

The refusals are the product

Six directional strategies were tested against real data. None cleared the bar. Each was stopped rather than tuned until it passed – which is the discipline the corpus exists to enforce.

STRATEGY	SOURCE	DATA	RESULT
VWAP regime filter	our screener	2,679 days	t = -1.29 – sign ran backwards
EMA / MACD / RelVol	teammate	32,148 bars	failed Bonferroni

STRATEGY	SOURCE	DATA	RESULT
20-day breakout	team	full history	$t = +1.08$ — not significant
Breakout + uptrend	refinement	full history	$t = -0.11$ — filter made it worse
News direction	research corpus	full history	$p = 0.44$ — coin flip
Crypto momentum	venue-wide	22 pairs	0 significant • median win 43.8%

An agent that explains why it declined 36 of 36 candidates demonstrates more than a P&L number can.

OUTLOOK

A distribution, not a prediction

We cannot forecast direction — five tested strategies established that. What we can state is where 136 comparable weeks, at today's volatility regime, have landed.

PERCENTILE	RESULT	% OF EQUITY	READING
p5	-2,374	-2.37%	very bad week
p25	+275	+0.27%	poor
p50 — MEDIAN	+2,144	+2.14%	typical
p75	+3,264	+3.26%	good
p95	+4,656	+4.66%	very good

WEEKS POSITIVE

80%

of 136 comparable weeks

RISK IS ASYMMETRIC

capped

upside limited by credit collected; downside by defined risk

DRAWS

1

the distribution describes many weeks — we get one

INFRASTRUCTURE

Everything is auditable

LEDGER

SHA-256

append-only, hash-chained, every evaluation

EXECUTION

2

independent switches required before any order

SCHEDULE

5 min

dry-run agent, cron, paths pinned

AUTONOMY

FULL

screens · gates · opens · closes · reconciles — unattended

REQUIREMENT	STATUS
Alpaca Trading API + CLI	all market data and execution
Options trading	iron condors — the core book
Autonomous execution	live 31 Aug · reconciles the book every cycle
Paper environment	PA3ID1B9L6BP · \$100,000
Public repository, MIT	pautax007/DELTAX
Live team board	pautax007.github.io/DELTAX

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